

Quick Check

Lesson 5: The Chemical Switch: How Juvenile Hormone Decides Whether an Insect Stays Young or Becomes an Adult

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.2: Growth and Development in Animals | Lesson 5

A learner says: "A caterpillar just melts inside the pupa and rebuilds itself without any control." Which fact best challenges this idea?

- A** Caterpillars of one species always become the same kind of butterfly, which suggests the changes are controlled by chemical signals in the body
- B** Caterpillars shed their skin, and the shed skin controls the next stage
- C** Pupae are always found on maize leaves, so the leaf controls the change
- D** Caterpillars stop eating when they become pupae, so nothing is happening inside

Which statement correctly describes the roles of the two hormones in insect growth?

- A** Ecdysone and JH both cause moulting, but neither affects the stage that follows
- B** Ecdysone triggers moulting and the level of JH decides the type of moult
- C** JH triggers moulting and ecdysone decides the type of moult
- D** JH is released by the prothoracic gland and it makes the insect stop moulting forever

A locust nymph in Turkana is about to moult. At this moult its Juvenile Hormone (JH) level is high. What will most likely result from the moult?

- A** A pupa
- B** No moult will take place because JH stops moulting
- C** A fully grown adult locust
- D** Another, larger nymph

A butterfly caterpillar has finished growing and its JH level starts to fall. What does the falling JH cause at the next moults?

- A** The caterpillar becomes a pupa, and later an adult butterfly
- B** The caterpillar changes straight into an adult without a pupa
- C** The caterpillar stops producing ecdysone and never moults again
- D** The caterpillar moults into a bigger caterpillar again and again

Why does a grasshopper have no pupa stage?

- A** Grasshopper nymphs have no exoskeleton, so they do not need a pupa
- B** Grasshoppers keep their JH level high for their whole life
- C** It has incomplete metamorphosis: when JH falls, the last nymph moults directly into an adult
- D** Grasshoppers do not make ecdysone, so they cannot moult into a pupa

Scientists add extra JH to a caterpillar in its final larval stage, when JH would normally fall. What is the most likely result?

- A** It becomes an adult butterfly at once
- B** It becomes a pupa earlier than normal
- C** It makes an extra larval stage, or a very large larva, instead of becoming a pupa
- D** Nothing changes, because only ecdysone affects the type of moult

A newly emerged grasshopper is larger than the empty exuvia (old skin) it left behind. Which explanation is correct?

- A** The hard exoskeleton stretches smoothly as the grasshopper grows
- B** The old skin peeled off like sunburnt skin and the grasshopper grew bigger afterwards
- C** The grasshopper's body expands before its new cuticle hardens
- D** The old skin shrank after it was shed

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).